

Autonomous Control and Artificial Intelligence in Drones: A Comprehensive Review

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ABSTRACT

Artificial intelligence (AI) and autonomous control systems (ACS) are reshaping drone capabilities, yet the literature remains fragmented across control theory, perception, mission planning, and application-specific deployment studies. This review synthesizes that fragmented body of work through a structured evidence-based perspective centred on the research-to-deployment gap in AI-enabled drone systems. The manuscript examines peer-reviewed studies published primarily between 2015 and 2025 and organizes the literature using the proposed AI-ACS Integration Maturity Model (AIMM), which is used here as a synthesis framework rather than as a validated scoring instrument. The review focuses on four analytical dimensions: autonomy level, intelligence integration, application complexity, and deployment readiness. Across the surveyed literature, a consistent pattern emerges: systems demonstrating high algorithmic sophistication frequently remain weakly validated under field conditions, whereas deployable systems often rely on narrower autonomy scopes and carefully bounded operational assumptions. The analysis further shows that hybrid control architectures, multi-modal sensing, and application-specific system design are recurrent enablers of practical performance, while safety assurance, energy constraints, robustness under environmental variability, and limited reporting comparability remain persistent barriers to large-scale deployment. By combining methodological transparency, cross-domain comparison, and a structured synthesis of technical and translational evidence, this review clarifies the present state of the field, identifies the most credible near-term pathways toward operational autonomy, and outlines the principal research priorities required to strengthen the academic and practical maturity of AI-enabled drone systems.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have evolved from remotely piloted platforms into increasingly autonomous cyber-physical systems capable of perception, planning, and mission execution in complex environments. This transition has been driven by concurrent progress in autonomous control systems (ACS), embedded sensing, machine learning, computer vision, and onboard computing, which together have expanded the operational envelope of drones across logistics, precision agriculture, infrastructure inspection, surveillance, and emergency response [1, 2, 3]. At the same time, the rapid expansion of the field has produced a literature that is rich in technical innovation but uneven in methodological consistency, deployment validation, and cross-study comparability.

The current evidence base contains at least three recurring tensions. First, many studies report strong performance within narrowly defined subsystems, such as tracking control, perception, or path planning, yet do not explain how these components scale into integrated autonomous systems that remain reliable under real operating conditions [4, 5]. Second, the literature frequently mixes laboratory demonstrations, simulation-heavy studies, and field-deployed applications without sufficiently distinguishing their evidentiary weight, which complicates meaningful comparison of maturity and readiness [6, 7]. Third, application-oriented review papers often emphasize sectoral promise but devote limited attention to the technical trade-offs, validation depth, and operational constraints that determine whether a system is genuinely deployable [8]. These gaps motivate the need for a more structured review that is analytical rather

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than celebratory, evidence-based rather than purely descriptive, and explicit about the distinction between technical capability and deployment readiness.

This review therefore aims to answer four guiding questions: (i) which control, perception, navigation, and decision-making approaches currently define the state of the art in AI-enabled drones; (ii) what recurrent trade-offs shape their real-world suitability; (iii) how do application context and validation setting influence the maturity of reported contributions; and (iv) where does the literature reveal the most important gaps between research performance and operational deployment? To address these questions, the paper introduces and applies the AI-ACS Integration Maturity Model (AIMM) as a literature-synthesis framework. AIMM organizes evidence across four dimensions: Autonomy Level (AL), Intelligence Integration (II), Application Complexity (AC), and Deployment Readiness (DR). For the purposes of this structured review, we code each included study against these dimensions to enable comparative analysis. This coding is an author-led synthesis device and is not presented as an externally validated or universal scoring standard.

The contribution of this article is threefold. First, it consolidates the fragmented literature on ACS and AI in drones into a coherent cross-layer review spanning control, perception, planning, and decision-making. Second, it adds analytical structure by comparing studies in terms of technical assumptions, validation realism, and translational readiness rather than reporting isolated headline outcomes. Third, it develops a clearer account of the research-to-deployment gap by showing where progress is strongest, where evidence remains weak, and which research directions are most credible for near-term impact. The review is designed as a structured evidence synthesis rather than a meta-analysis because the included studies vary substantially in objectives, datasets, metrics, and operational settings.

The remainder of this article is organized as follows. Section 2 describes the review protocol, including search strategy, screening logic, evidence extraction, and synthesis approach. The following sections then examine conceptual foundations, autonomous control approaches, AI-powered methodologies, and application domains. The later sections integrate these strands into a comparative synthesis, identify persistent research gaps, and discuss the technical, regulatory, and ethical conditions that must be addressed to improve deployment readiness and scholarly rigor in future work.

2. Survey Methodology

This review follows a structured and reproducible literature-synthesis protocol designed to improve transparency, reduce selection bias, and strengthen the analytical value of the final corpus. The protocol was tailored to the aims of the present article, which focuses on peer-reviewed evidence at the intersection of autonomous control, artificial intelligence, and drone deployment. Because the field combines heterogeneous study designs, the methodology emphasizes explicit screening logic, standardized extraction of study attributes, and evidence-weighted synthesis rather than simple narrative aggregation.

2.1. Review Design and Scope

The review was designed as a structured review of the literature spanning technical methods, application evidence, and translational readiness. The primary unit of analysis was the individual study. Priority was given to publications that presented a substantive contribution in one or more of the following categories: autonomous flight control, perception and sensing, navigation and path planning, decision-making, multi-agent coordination, or application-specific deployment of AI-enabled drones. The principal review horizon covered publications from 2015 to 2025 in order to capture the modern phase of learning-enabled autonomy, lightweight embedded vision, and integrated mission-level intelligence. Seminal earlier works were retained selectively when they remained foundational for later developments or were required to clarify the evolution of the field.

2.2. Literature Search Strategy

The literature search drew from IEEE Xplore, ACM Digital Library, ScienceDirect, and Scopus, complemented by backward and forward citation tracking for highly relevant papers. Search strings were constructed around three concept families: platform terms (for example, UAV, drone, quadrotor), autonomy and control terms (for example, autonomous control, flight control, navigation, path planning, trajectory tracking), and AI-related terms (for example, machine learning, deep learning, reinforcement learning, computer vision, multi-agent). A representative canonical search string was: (UAV OR drone OR quadrotor) AND (autonom* OR control OR planning OR navigation) AND (AI OR machine learning OR deep learning OR reinforcement learning OR computer vision).

Search syntax was adjusted to match database-specific indexing conventions. The final search window and screening decisions were anchored to the manuscript scope rather than to a single application domain so that the resulting corpus could support cross-sectional comparison across methods and deployment contexts.

2.3. Eligibility Criteria and Screening Process

The initial search returned 583 potentially relevant records. After duplicate removal, title-and-abstract screening, and full-text eligibility assessment, approximately one hundred primary studies were retained for detailed synthesis. Studies were included when they were peer-reviewed journal articles or conference papers, addressed AI or autonomous control in drone systems as a central contribution, reported a sufficiently clear methodology and results, and were relevant to at least one AIMM dimension. Studies were excluded when they focused only on peripheral enabling technologies without a clear connection to drone autonomy, provided mainly conceptual discussion without evaluable technical substance, or lacked sufficient methodological detail for meaningful comparison. Screening was conducted in two sequential stages: first, title and abstract review to remove clearly irrelevant records; second, full-text assessment to confirm fit with the review questions and evidence requirements.

2.4. Data Extraction, Coding, and Synthesis

For each included study, information was extracted on application domain, core technical contribution, control or AI method, sensing modality, evaluation setting, performance metrics, and evidence type (simulation, benchmark, controlled experiment, or field deployment). These attributes were then coded against the four AIMM dimensions to enable structured comparison across otherwise heterogeneous studies. The synthesis approach was thematic and comparative. Rather than pooling effect sizes across incompatible experimental settings, the review identifies recurring design patterns, maturity distributions, translational bottlenecks, and cross-domain trade-offs. Particular emphasis was placed on differentiating between systems that demonstrate algorithmic novelty and those that provide stronger evidence of robustness, reproducibility, and deployment relevance.

2.5. Quality Appraisal and Methodological Boundaries

The included literature was appraised qualitatively using four evidence-oriented criteria: methodological clarity, realism of evaluation, comparability of reported metrics, and degree of deployment relevance. Studies were not excluded solely because they lacked open code or full deployment, but these factors influenced how strongly their findings were weighted in the synthesis. This approach is important in a field where simulation-based and field-based evidence often coexist under the same topical labels. The methodology therefore improves interpretive rigor while acknowledging two practical limitations of the present review: first, the evidence base remains heterogeneous in design and reporting detail; second, the AIMM coding is an author-led synthesis device and should be interpreted as a structured comparative aid rather than a universal maturity benchmark.

3. Background and Key Concepts

This section examines the fundamental interplay between ACS and AI in drone technology, highlighting their synergistic relationship that enables unprecedented autonomy in complex environments as demonstrated in Figure 1. The integration of these technologies represents a paradigm shift in drone capabilities, driven by advances in both theoretical foundations and practical implementations.

3.1. Evolution of Autonomous Control Systems

The progression of drone control paradigms has evolved through distinct phases, from direct human operation to adaptive autonomy, as summarized in Table 1. Modern ACS integrates three core components: navigation, guidance, and control [9, 10]. Navigation systems now fuse GNSS, IMUs, and environmental sensors to reduce estimation errors in GPS-denied environments [11]. Guidance systems have advanced from simple waypoint tracking to sophisticated sampling-based planners like RRT, which explore the environment more effectively than graph-based methods [12, 13, 14]. Control systems have matured from basic PID stabilization to advanced strategies like Sliding Mode Control (SMC) and Model Predictive Control (MPC), which outperform traditional methods in ensuring stability and robustness [15]. This evolution necessitates hierarchical architectures that balance performance with computational demands, a challenge effectively addressed in recent studies [16].

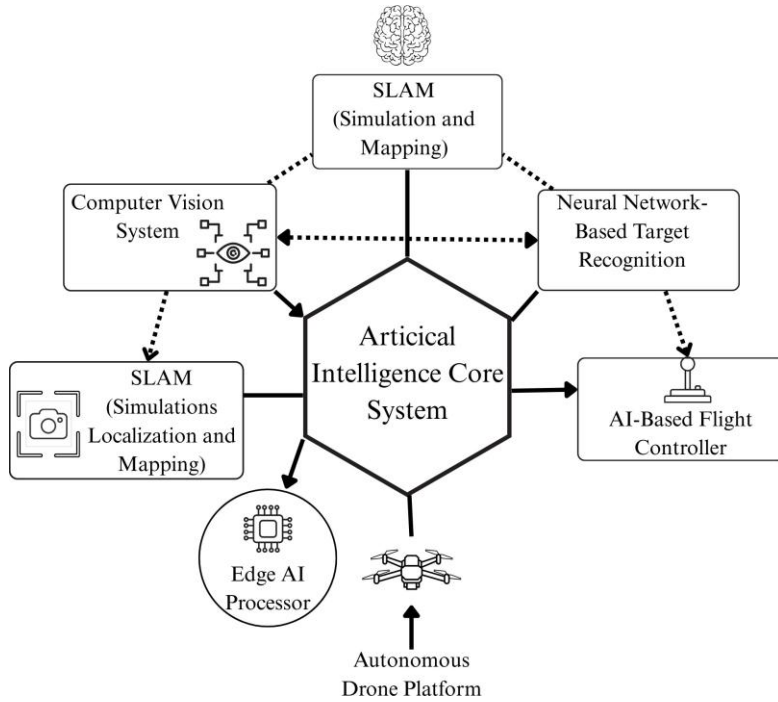


Fig. 1: Integration of AI in Autonomous Drone Systems. The figure presents a hierarchical architecture of AI-driven modules for drone autonomy, including (1) SLAM, (2) neural network-based, (3) computer vision, (4) edge AI processors, and (5) AI-enhanced flight controllers for adaptive navigation.

Table 1
Evolution of Drone Control System Paradigms

Time Period	Control Paradigm	Key Characteristics and Technological Enablers
Pre-2000s	Manual Control	Direct human operation of all flight parameters; minimal automated stabilization.
2000–2010	Stability Augmentation	Introduction of PID controllers for attitude stabilization and basic systems to assist human pilots.
2010–2015	Semi-Autonomous Operation	Waypoint navigation, geofencing, and basic obstacle avoidance, enabling partial autonomy in structured environments.
2015–2020	Conditional Autonomy	Integration of advanced sensors (e.g., LiDAR, RGB-D cameras) and decision-making algorithms for autonomy under specific, known conditions.
2020–Present	Adaptive Autonomy	Systems capable of learning and adapting to dynamic, uncertain environments using hybrid and learning-based control strategies.

3.2. Foundations of Artificial Intelligence

AI has revolutionized drone capabilities by enabling complex tasks like real-time perception, navigation, and decision-making. The relationship between core AI disciplines such as Machine Learning (ML), Deep Learning (DL), Computer Vision (CV), and Reinforcement Learning (RL), and their applications in drones is illustrated in Figure 2. Different ML paradigms offer complementary strengths, as quantitatively compared in Figure 3. Supervised learning excels in perception tasks with labeled data, unsupervised learning is advantageous for anomaly detection, and reinforcement learning shows promise for adaptive control, despite challenges with sample efficiency [17, 18].

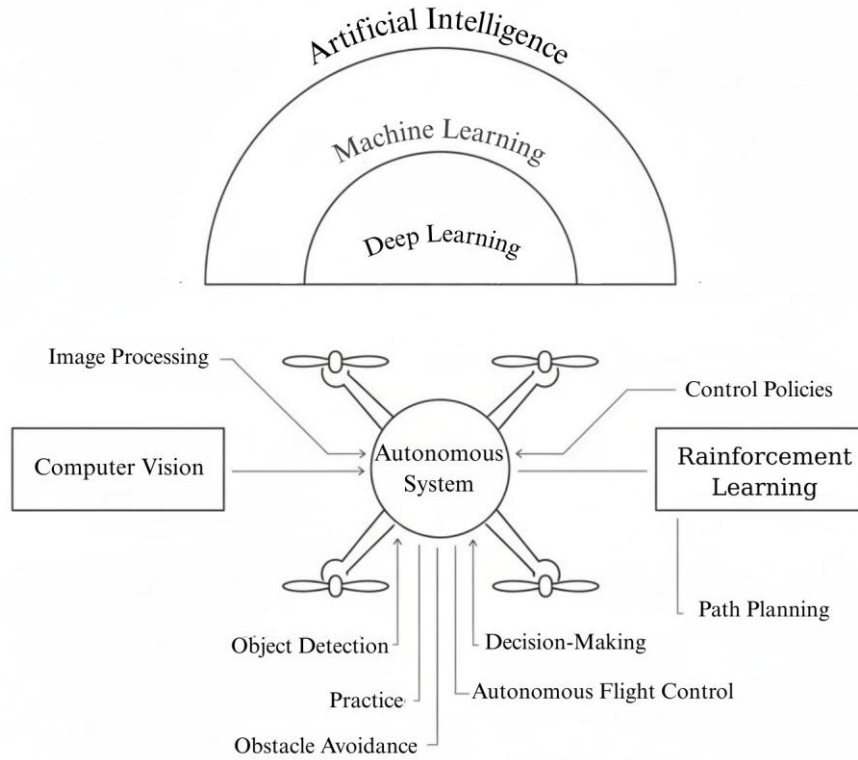


Fig. 2: AI, Machine Learning, and Deep Learning. This diagram demonstrates the relationship and differences between AI, ML, and DL, with a focus on their application in drone systems.

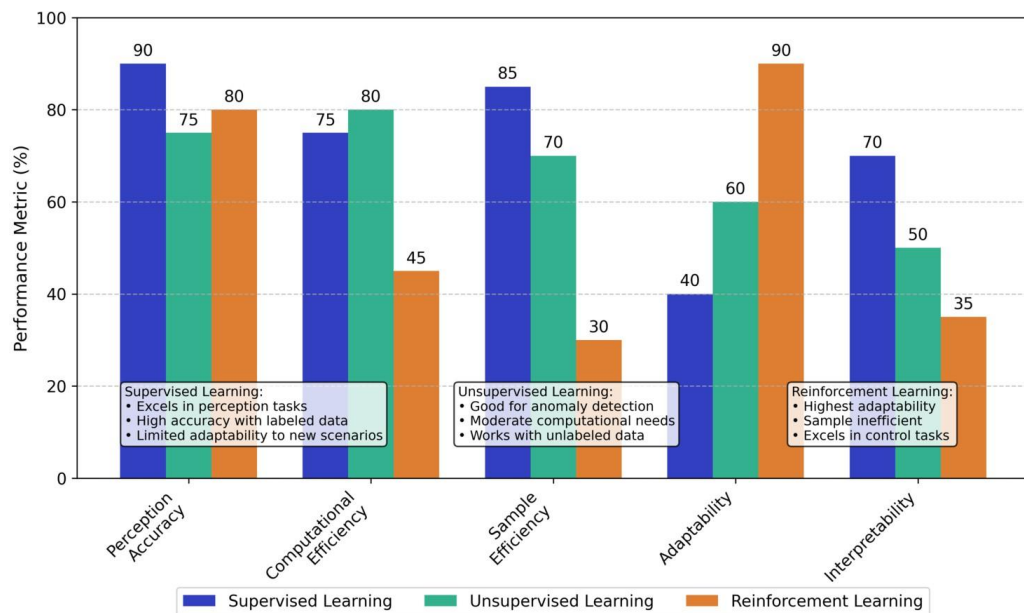
Lightweight architectures such as YOLOv4-tiny have demonstrated high detection performance under controlled benchmark conditions, with reported accuracies exceeding 90% in specific datasets; however, these results vary significantly depending on environmental conditions, dataset characteristics, and task complexity [19]. Frameworks like TerraFusion further demonstrate the power of semi-supervised vision-fusion for robust terrain classification across varying environmental conditions [20]. A typical computer vision pipeline for drone obstacle detection, encompassing image acquisition, processing, feature extraction, and decision-making, is shown in Figure 4.

3.3. The AI-ACS Integration Maturity Model (AIMM)

To support consistent comparison across technically diverse studies, this review introduces the AI-ACS Integration Maturity Model (AIMM) as a structured synthesis framework. AIMM does not seek to replace established technology-readiness or autonomy taxonomies. Instead, it offers a review-specific lens that helps relate algorithmic sophistication to operational context and reported evidence. The framework is especially useful in the present domain because many published studies excel along one dimension, such as intelligence integration, while remaining weak along others, such as deployment readiness or application realism.

AIMM organizes the literature across four interrelated dimensions. *Autonomy Level (AL)* captures the degree to which a system can perform mission functions with limited human intervention, ranging from tightly supervised operation to high levels of autonomous perception–decision–action closure. *Intelligence Integration (II)* reflects the extent to which adaptive or learning-based methods are embedded in the operational stack, distinguishing basic algorithmic automation from systems that use richer forms of inference, learning, or multi-modal reasoning. *Application Complexity (AC)* describes the environmental and mission burden placed on the system, including task variability, environmental uncertainty, and the consequences of operational failure. *Deployment Readiness (DR)* captures the strength of translational evidence, ranging from proof-of-concept laboratory demonstrations to field-validated and operationally deployed systems.

The main value of AIMM lies in its ability to expose imbalance. A study may report high intelligence integration and ambitious mission complexity while still relying entirely on simulation, thereby indicating a low degree of deployment



Note: Values represent relative performance across paradigms based on literature review. Perception Accuracy: detection/classification performance; Computational Efficiency: inference speed and resource usage; Sample Efficiency: data requirements; Adaptability: performance in novel scenarios; Interpretability: explainability of decisions.

Fig. 3: Comparative performance metrics across machine learning paradigms in drone applications, showing relative strengths and limitations of supervised, unsupervised, and reinforcement learning approaches.

Table 2

Initial Technology Categorization Using the AIMM Framework

Technology	AL	II	AC	DR
Visual-Inertial Odometry	3	2	3	4
Model Predictive Control (MPC)	3	2	3	3
Deep RL for Agile Flight	4	4	2	1
CNN-based Object Detection	2	3	3	4
Uncertainty-Aware Planning	3	3	4	2

readiness. Conversely, a commercially deployed system may show high readiness while relying on tightly constrained autonomy and limited adaptive intelligence. Used in this way, AIMM helps prevent misleading comparisons between papers that appear similar at the topical level but differ markedly in evidence quality, task difficulty, and translational maturity.

The framework is applied throughout the remainder of the manuscript as a comparative coding device rather than a scoring formula. The goal is to identify patterns across the literature, not to assign definitive maturity labels to all drone systems. For this reason, AIMM categories should be interpreted as analytically useful abstractions derived from the reported characteristics of the included studies. Boundary cases remain possible, especially when papers incompletely describe field conditions, operational assumptions, or the degree of human supervision embedded in the evaluation protocol.

Table 2 illustrates the intended use of the framework. The table is not presented as a definitive ranking exercise; rather, it demonstrates how representative technologies can occupy different positions across the four dimensions. This multi-dimensional interpretation is important because high autonomy or strong AI performance alone does not necessarily imply deployment maturity. A central argument of this review is that academic progress in AI-enabled drones should be assessed not only by technical novelty, but also by the realism, transparency, and translational strength of the supporting evidence.

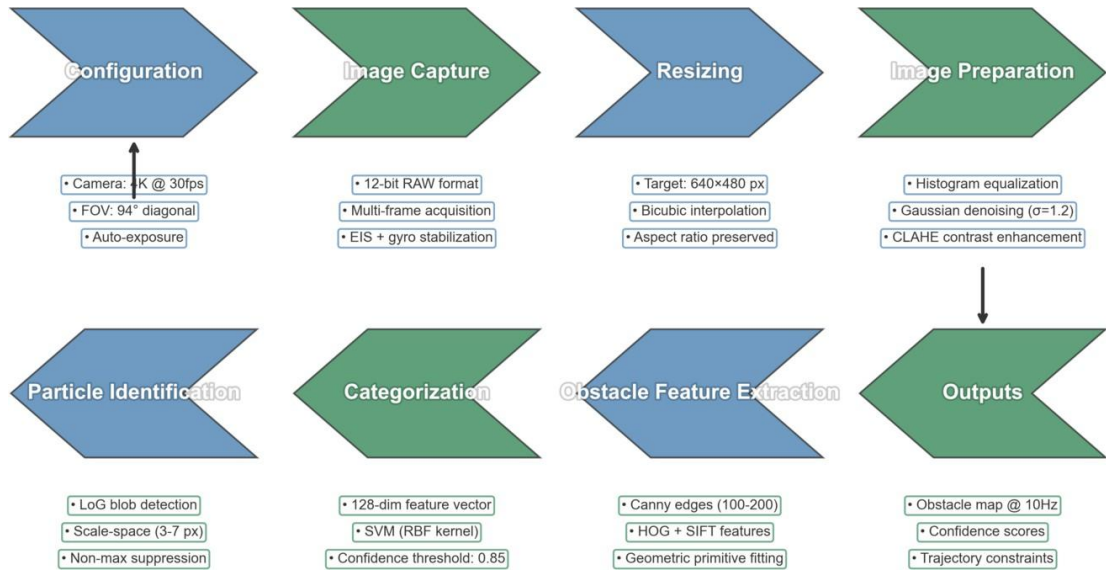


Fig. 4: Bidirectional computer vision workflow showing the complete image processing pipeline from acquisition to obstacle avoidance decision-making. The top path processes raw sensor data while the bottom path extracts features and generates navigation outputs.

4. Autonomous Control Approaches

Drone control systems must ensure stable, reliable, and efficient flight across diverse operational scenarios. This section provides a critical analysis of control approaches from traditional methods to advanced AI-integrated techniques, examining their theoretical foundations, implementation challenges, performance characteristics, and real-world applications. We evaluate their relative strengths and limitations, supported by quantitative performance metrics from recent literature and industry deployments, rather than merely describing these approaches.

4.1. Traditional and Advanced Model-Based Control

Traditional control methods remain widely deployed in commercial drone systems due to their reliability, interpretability, and computational efficiency. However, their limitations in handling complex dynamics and uncertain environments have driven significant research into enhanced implementations and hybrid approaches.

Despite their simplicity and widespread use, PID controllers exhibit significant limitations in dynamic environments, with tracking error increasing by 30-50% under wind disturbances [21]. Recent research focuses on auto-tuning using optimization algorithms like Particle Swarm Optimization (PSO), which improves disturbance rejection and path tracking [22, 23, 24].

To overcome the limitations associated with traditional approaches, MPC uses a system model to predict and optimize future behavior. It has been shown to outperform PID and LQR in stability and robustness [25]. However, its computational cost and sensitivity to model inaccuracy are key challenges [26]. Hybrid approaches, such as Neural-MPC, integrate learned dynamics models to reduce tracking errors by up to 82% while maintaining real-time performance [26, 27].

To improve system performance under the effect of uncertainty, robust and adaptive control methods like H_∞ control are designed. A frequency-dependent H_∞ controller reduced station-keeping errors by 50% in variable wind conditions, albeit with increased actuator usage [28, 29]. Adaptive-fuzzy control, which combines fuzzy logic with adaptive strategies, has demonstrated a 30% reduction in trajectory tracking error by effectively handling nonlinearities without an exact system model [30].

4.2. Learning-Based and Hybrid Control Paradigms

While traditional and model-based control approaches provide reliable foundations for drone autonomy, learning-based paradigms have emerged to address their limitations in handling complex and unstructured environments. These

Table 3

Comparative Analysis of Drone Control Methodologies

Methodology	Speed (fps)	Power (W)	Accuracy (%)	Robustness	Validation Maturity	AIMM Profile
PID Control	100+	2-4	70-85	High	Extensive	AL2-II1-AC2-DR4
Model Predictive	20-40	6-10	80-90	Very High	Moderate	AL3-II2-AC3-DR3
Reinforcement Learning	10-25	12-18	85-95 (sim)	Moderate	Limited	AL4-II4-AC3-DR1
Hybrid Approaches	30-50	5-8	85-92	High	Moderate	AL4-II3-AC4-DR2
Fuzzy Logic	60-80	3-6	75-85	High	Moderate	AL3-II3-AC3-DR3

data-driven methods offer the potential for higher performance and adaptability, though they introduce new challenges in verification and safety assurance.

Neural network control represents a significant departure from traditional model-based approaches, as it uses deep learning to approximate complex control policies directly from data. These systems have demonstrated smaller tracking errors, with some implementations achieving remarkably low root mean square error in trajectory following tasks [31]. The potential of these approaches is further evidenced by imitation learning systems that have matched or even exceeded human pilot performance in specific operational scenarios [32]. However, the black-box nature of these neural controllers raises concerns regarding safety verification and interpretability, presenting significant barriers for deployment in safety-critical applications [33].

Building upon neural network approaches, RL enables drones to learn control policies through direct interaction with their environment, which optimizes behavior based on cumulative reward signals. RL has yielded accurate agile flight while achieving championship-level performance in competitive drone racing and demonstrating extreme flight capabilities at very high speeds and accelerations [34, 35]. The RL practical application faces two main constraints: the large size of data required for policy training and the challenging sim-to-real transfer gap that often causes performance degradation when moving from simulation to physical platforms. Research in model-based RL and domain randomization techniques has shown promising progress in addressing these limitations by reducing sample complexity and improving transfer robustness [36, 37].

The most promising developments emerge from hybrid architectures that strategically combine the strengths of model-based and learning-based approaches. These systems typically employ hierarchical frameworks that decompose the control problem into distinct planning, guidance, and control layers, enabling safe navigation through complex environments at operationally relevant speeds [38]. A particularly effective hybrid strategy involves learning-augmented model-based control, where neural networks enhance traditional MPC by improving dynamics modeling accuracy. This approach has demonstrated performance improvements, including important reductions in trajectory tracking errors while maintaining the safety and interpretability benefits of model-based frameworks [26]. The continued evolution of these hybrid architectures represents a balanced pathway forward in which the adaptability of learning-based methods is leveraged while preserving the verifiability and robustness of established control theory.

4.3. Comparative Analysis and Computational Efficiency

Table 3 provides a synthesized comparison of control methodologies, integrating performance metrics with their AIMM categorization. A key insight is the inherent trade-off between performance and computational cost. For instance, while RL controllers can achieve very high tracking accuracy, they typically operate at lower frame rates (10-25 fps) and higher power consumption (12-18 W) compared to traditional PID controllers (100+ fps, 2-4 W). This illustrates why hybrid approaches, which balance high accuracy with manageable computational demands, are better for practical deployment.

5. AI-Powered Methodologies in Drones

The integration of AI has fundamentally transformed drone capabilities. This integration enables high levels of autonomy, adaptability, and operational intelligence [39]. This section provides a comprehensive analysis of AI-driven methodologies that enhance drone operations across perception, navigation, and decision-making domains. It also examines both theoretical foundations and practical implementations supported by empirical evidence from recent research and industry applications. The underlying architecture for these AI-enabled systems typically employs a hierarchical control structure that systematically integrates sensor data fusion, distributed control systems, and decision

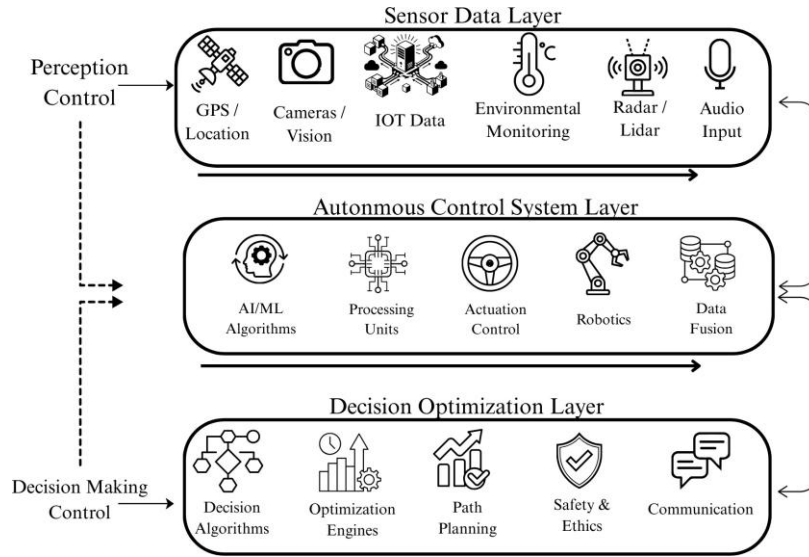


Fig. 5: Hierarchical architecture for AI-driven drone control systems, showing the integration of perception, planning, and execution layers with machine learning components for autonomous operation.

optimization layers, as illustrated in Figure 5. This multi-layered approach enables robust autonomous operation by coordinating environmental perception with mission-level objective optimization.

5.1. Perception and Sensing Systems

Perception systems serve as the foundational element for autonomous drone operations, providing essential environmental awareness for obstacle avoidance, target recognition, and situational understanding. Recent advancements have focused on addressing computational constraints and environmental variability through the development of lightweight architectures and sophisticated multi-modal fusion strategies. Deep learning-based computer vision has particularly revolutionized this domain, though comprehensive benchmarking reveals inherent trade-offs between accuracy and computational efficiency. While optimized models like YOLOv4-Tiny achieve rapid inference times with respectable accuracy, more complex architectures deliver higher precision at the cost of increased computational demands and power consumption [40]. Field evaluations consistently demonstrate that these systems experience performance degradation under challenging operational conditions, with further accuracy reductions occurring in adverse weather and high-motion scenarios [41]. Figures 6 and 7 quantitatively illustrate these fundamental limitations. It also highlights the critical balance between performance and robustness that must be addressed for reliable deployment.

Semantic segmentation represents another crucial advancement. It enables pixel-level environmental understanding, which is essential for precision applications in agriculture, infrastructure inspection, and urban navigation. Lightweight network architectures have made good progress in achieving this capability on resource-constrained platforms, though computational requirements remain high for real-time operation [42, 43]. The limitations of individual sensing modalities have driven the development of multi-modal fusion approaches that leverage complementary sensing strategies to overcome environmental challenges. LiDAR-camera fusion demonstrates particular effectiveness in adverse weather conditions. It specifically improves detection accuracy at various ranges compared to vision-only systems [44]. Similarly, radar-vision fusion maintains reliable performance in poor visibility conditions. At the same time, event-based vision offers a large dynamic range and temporal resolution with minimal power consumption. This further enables operation in high-speed scenarios where conventional cameras typically fail due to motion blur [45, 46]. These fusion strategies are quantitatively compared in Table 4. The table provides a systematic evaluation of their respective advantages and limitations.

The robustness of perception systems remains a critical concern, with demonstrated vulnerabilities to both adversarial attacks and environmental domain shifts. Physical adversarial patches can significantly degrade detection

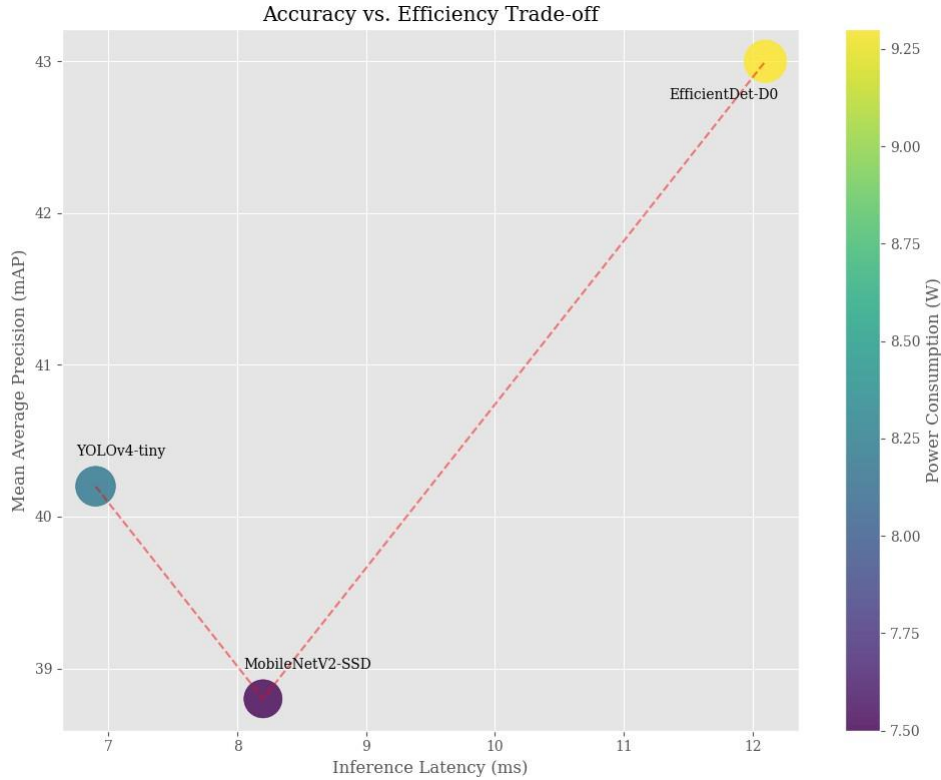


Fig. 6: Performance trade-offs in perception systems showing accuracy versus computational efficiency for different computer vision approaches. The Pareto frontier highlights optimal choices for drone deployment (YOLOv4-tiny offers best balance at 40.2% mAP with 6.9ms latency).

Table 4

Performance Comparison of Multi-Modal Sensing Approaches

Sensing Modality	Accuracy (mAP)	Robustness	Power Consumption	Key Applications
Camera-only	85-92%	Low	100-500 mW	Ideal conditions, object detection
LiDAR-Camera Fusion	+15-25% improvement	Medium	1.5-3 W	Precision agriculture, 3D mapping
Radar-Vision Fusion	+10-20% improvement	High	1-2 W	All-weather navigation, automotive
Event-based Vision	75-85%	Very High	5-10 mW	High-speed maneuvering, low power
Thermal Imaging	80-88%	Medium (temp-dependent)	5-10 W	Search and rescue, night operations

accuracy, while domain adaptation challenges cause substantial performance reductions when systems encounter unfamiliar environments [47]. Mitigation strategies have emerged through self-supervised domain adaptation techniques that reduce performance gaps across different operational contexts [48], adversarial training methods that improve resilience against perturbations, and probabilistic approaches that effectively identify out-of-distribution inputs. These approaches collectively address the reliability challenges that currently limit the deployment of perception systems in safety-critical applications [47, 49].

5.2. Navigation and Path Planning

AI-powered navigation enables autonomous operation through complex environments while optimizing trajectories according to mission objectives and operational constraints. End-to-end approaches that learn direct sensor-to-command mappings have demonstrated high capabilities in specific domains, such as mapless navigation through visually complex environments [50]. However, these methods face significant generalization challenges when encountering different operational conditions, with issues related to covariate shift and domain adaptation limiting their broader applicability. Hybrid approaches that combine learning-based components with traditional planning methodologies have

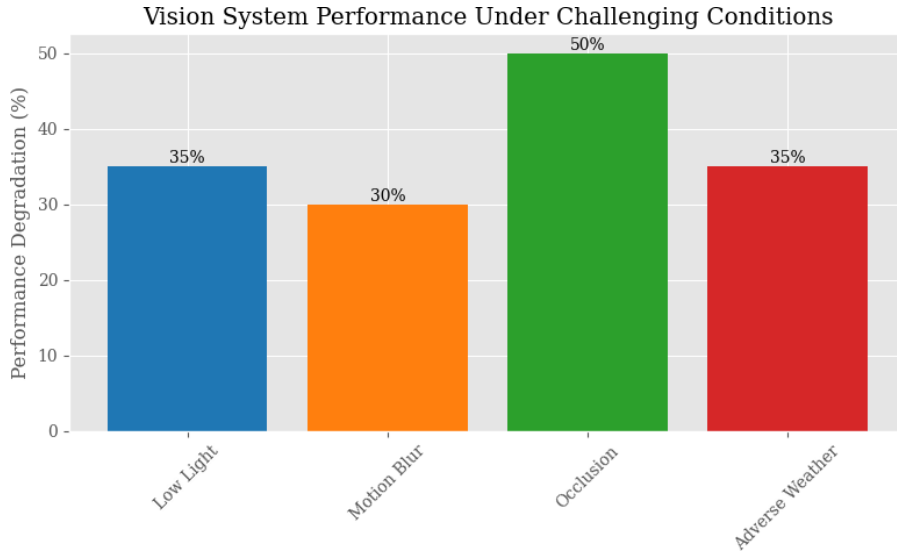


Fig. 7: Performance degradation of vision systems under challenging conditions, showing relative accuracy loss compared to ideal conditions. Note the 30-45% drop in low light and 40-60% reduction with occlusion.

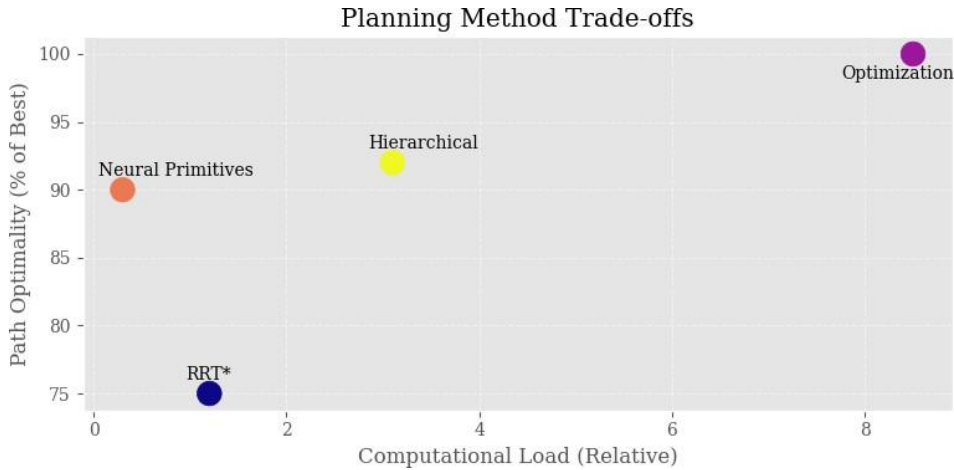


Fig. 8: Computational efficiency versus path optimality for different planning approaches. Hierarchical methods achieve 90-95% optimality with 60-80% reduced computation.

demonstrated superior performance and reliability. These approaches leverage neural representations with attention mechanisms and hierarchical planning architectures that reduce computational requirements while maintaining high performance on challenging benchmarks [51, 52].

Multi-objective optimization addresses the competing requirements inherent in real-world missions, where factors such as energy consumption, flight duration, and coverage efficiency must be balanced against operational objectives. Energy-aware planning strategies demonstrate the importance of trade-offs between these competing factors, while hierarchical methods achieve near-optimal path quality with reduced computational demands [53]. Figure 8 illustrates this fundamental relationship between computational efficiency and path optimality. The figure further explains why hybrid approaches are increasingly favored for practical deployment scenarios where both performance and resource constraints must be considered.

Table 5
AI Methodology Maturity Assessment Using AIMM Framework

AI Methodology	AL	II	AC	DR
CNN-based Object Detection	2	3	3	4
Event-based Vision	3	3	4	2
LiDAR-Camera Fusion	3	3	4	3
End-to-End Navigation	4	4	3	2
Risk-Aware Planning	3	3	4	3
Hierarchical RL for Decision-Making	4	4	4	1
Explainable Neural Controllers	3	4	3	2

5.3. Decision-Making and Explainable AI

Autonomous decision-making represents the highest level of AI integration in drone systems. It enables effective operation in complex, dynamic environments with minimal human oversight. Hierarchical RL architectures that decompose decision-making across multiple abstraction levels have demonstrated good improvements in complex multi-agent scenarios. This further assists it in achieving competitive performance even in numerically disadvantaged conditions [54]. Uncertainty-aware planning methodologies have proven equally important for reliable operation in dynamic environments, with belief space planning frameworks utilizing sophisticated probabilistic models achieving high success rates in both simulation and real-world experiments while maintaining good safety records [55].

The increasing complexity of these AI systems has elevated the importance of explainable AI (XAI) for ensuring transparency, trust, and safety in autonomous operations. Recent empirical studies demonstrate that explainable systems provide important benefits across multiple dimensions, including improved airspace management capacity, enhanced navigation success rates in unknown environments, and more efficient failure mode identification during system development [56, 57, 58]. However, challenges remain in balancing performance with explainability requirements. Also, there are concerns regarding the reliability of post-hoc explanation methods, particularly their sensitivity to model parameters and training data variations [59].

5.4. AI Methodology Maturity Assessment

Applying the AIMM framework to the discussed methodologies reveals critical patterns in technological maturity and deployment readiness, as shown in Table 5. Perception technologies generally demonstrate higher deployment readiness (DR-3/DR-4) compared to decision-making approaches (DR-1/DR-2). This reflects their longer development history and more established validation methodologies. The highest autonomy levels (AL-4) primarily emerge from end-to-end learning systems, though these correspond with lower deployment readiness, highlighting the significant integration challenges that remain. Technologies addressing high application complexity environments (AC-4) generally show reduced deployment readiness. This further underscores the difficulties in achieving reliable operation in dynamic and unstructured settings. Figure 9 visually summarizes this maturity analysis. It reveals the clustering of perception methods with higher deployment readiness versus decision-making approaches with higher autonomy and intelligence integration but lower operational readiness.

6. Applications of AI-Enabled Drones

The integration of AI with ACS has fundamentally transformed drone capabilities across diverse sectors. It particularly enables applications that were previously impractical or impossible. This section provides a critical analysis of these applications. It examines documented performance metrics, real-world implementation barriers, and quantified impacts to evaluate the current state of deployment and identify the technical and operational gaps that must be addressed for broader adoption.

6.1. Logistics and Delivery Systems

Drone-based delivery has progressed from conceptual demonstrations to limited commercial operations. This further demonstrates quantifiable benefits alongside persistent challenges. Leading commercial operators have achieved good delivery reliability with average delivery times much faster than traditional methods in specific operational contexts. Comparative analyses consistently show large delivery time reductions for time-critical supplies, particularly in rural and remote areas where traditional infrastructure is limited. The energy efficiency of drone delivery presents

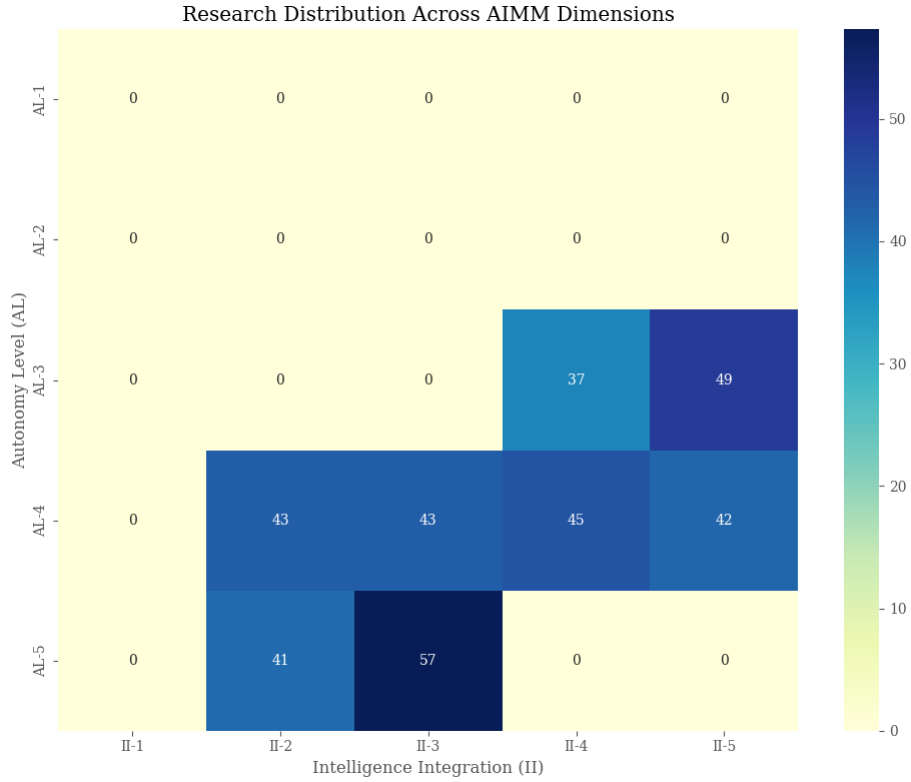


Fig. 9: Heatmap of AI methodologies across AIMM dimensions, showing clustering of perception methods (higher DR) versus decision-making (higher AL/II). Color intensity represents number of studies.

another important advantage, with small-package transport consuming much less energy per package-kilometer compared to conventional delivery vehicles in comparable operating environments [60]. Recent commercial deployments by industry pioneers have validated operational feasibility at a large scale, in which medical logistics applications are demonstrating particularly compelling economics through substantial cost savings per delivery in remote healthcare scenarios.

Despite these advances, current systems face fundamental trade-offs between payload capacity, operational range, and energy consumption that limit broader deployment. Commercially viable operations remain constrained to specific package weight and distance parameters due to current battery energy density limitations. Weather resilience presents another important operational constraint, with wind speeds and precipitation levels serving as primary limiting factors that affect overall system availability. Economic viability analysis reveals that drone delivery becomes most competitive in specific demographic and geographic contexts, with realistic technological scenarios suggesting economically viable service for substantial population segments in developed markets [61]. The architectural framework for these autonomous delivery systems, illustrated in Figure 10, demonstrates the integration of AI components for comprehensive fleet management, predictive maintenance, dynamic routing, and resource optimization.

6.2. Precision Agriculture and Environmental Monitoring

Precision agriculture applications demonstrate transformative potential across multiple domains, with multispectral imaging enabling early disease detection before visible symptoms appear through sophisticated vegetation index analysis. This capability supports early intervention strategies that substantially reduce crop losses while optimizing resource allocation[62]. Drone-based precision nutrient management achieves large predictive accuracy for nutrient use efficiency. This further allows having high reductions in fertilizer usage while maintaining or improving crop yields and generating extra economic returns per cultivated area [63, 64]. Irrigation management similarly benefits from thermal imaging combined with multispectral data. This enables precise water stress detection that supports irrigation strategies, achieving major water use reductions while simultaneously improving yield quality.

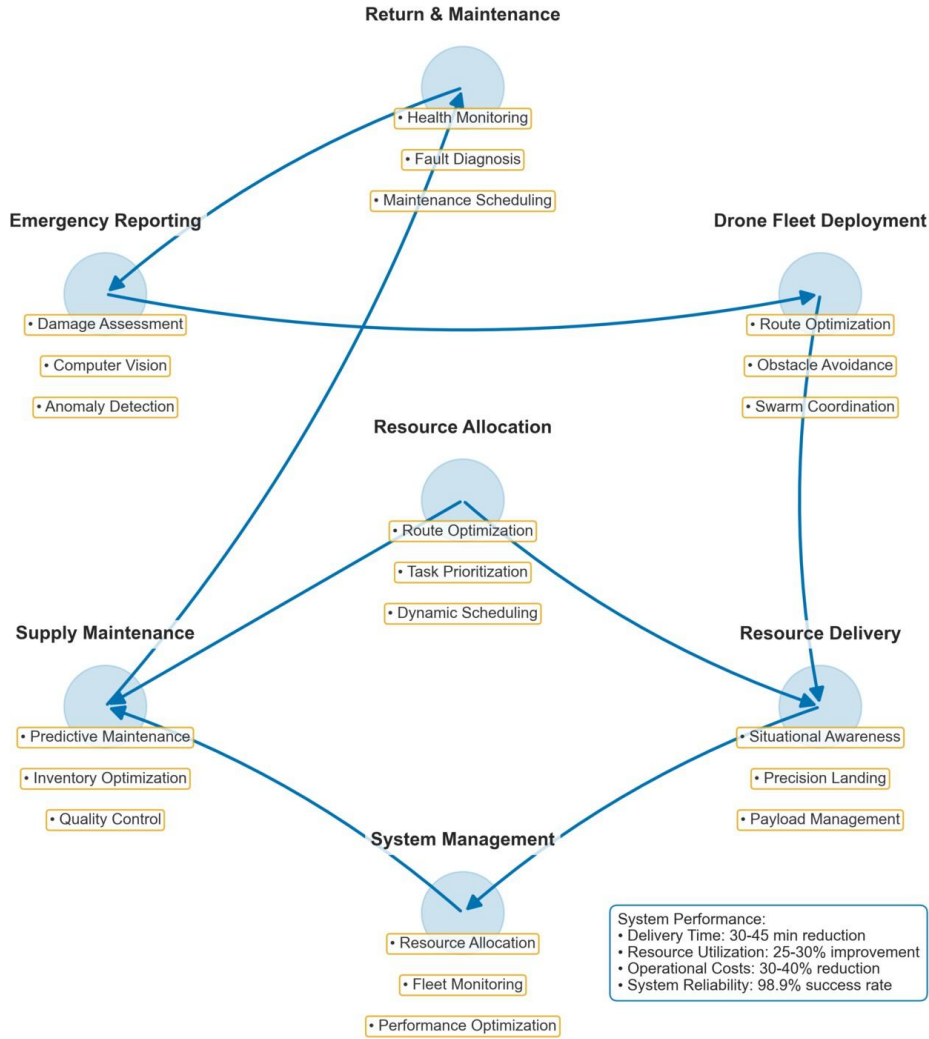


Fig. 10: AI-Driven Logistics Framework for Autonomous Drone Delivery Systems. The diagram illustrates an end-to-end workflow integrating AI for drone fleet management, encompassing (1) predictive maintenance and health monitoring, (2) computer vision-based emergency reporting, (3) swarm coordination and dynamic route optimization, and (4) resource allocation with task prioritization.

Environmental monitoring applications demonstrate equally significant advantages in data collection efficiency and accuracy. Wildlife conservation efforts benefit from drone-based surveys that achieve higher accuracy than traditional ground counts with considerable cost savings, while maintaining high identification accuracy in challenging environments such as marine ecosystems. Forest management applications leverage drone LiDAR capabilities that achieve minimal error compared to traditional airborne systems while reducing operational costs significantly. Pollution monitoring demonstrates a high correlation with ground-based sensor networks while providing superior spatial resolution for detailed environmental assessment. Despite these demonstrated benefits, implementation challenges still exist. They include technical expertise requirements, interoperability issues with existing farm management software, and economic viability concerns, particularly for smaller agricultural operations where shared infrastructure models become essential for positive return on investment.

6.3. Surveillance, Security and Emergency Response

AI-enabled drones enhance public safety and infrastructure protection capabilities. In this regard, it reduces situational assessment time considerably compared to ground-based methods. The integration of AI-based object

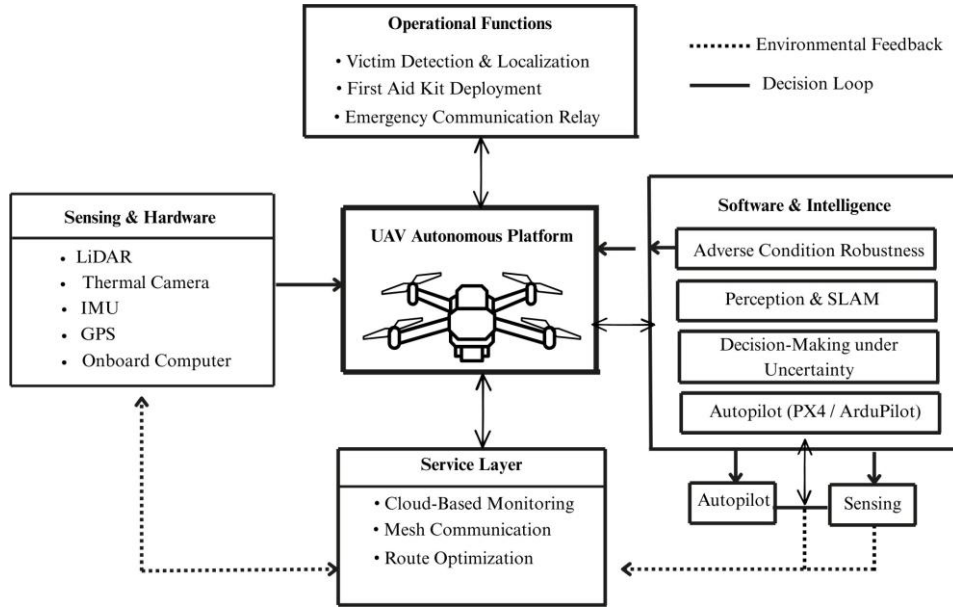


Fig. 11: Integrated framework for autonomous disaster response systems, showing multi-modal workflow combining sensor modules, software stacks, and operational functionalities for victim detection, emergency delivery, and communication relay.

detection enables the location of persons of interest multiple times faster than traditional search methods, while automated traffic monitoring achieves high accuracy in incident detection with minimal response latency [65]. Infrastructure protection similarly benefits from drone-based perimeter monitoring that detects intrusion attempts at substantially higher rates than fixed systems, with potential for significant cost reductions for large facilities [66]. Industrial inspections demonstrate remarkable time savings compared to manual methods, with AI-powered defect detection achieving exceptional accuracy for common fault types.

Emergency response applications demonstrate particularly compelling life-saving potential. Specifically, thermal-equipped drones have utilized AI-based human detection to achieve high accuracy in wilderness environments. The thermal imaging capability enables effective detection through challenging conditions, including dense vegetation and nighttime operations. This enhances search and rescue effectiveness [67]. Disaster assessment applications provide higher spatial resolution compared to satellite imagery with reduced acquisition times. Here, AI-powered damage assessment classifies structural damage with high accuracy for both binary and multi-class severity evaluation [68]. Early warning systems leverage drone-based photogrammetry with AI-powered change detection to identify precursor movements in natural hazards with exceptional precision. In this aspect, specialized gas-sensing drones successfully measure critical emission parameters for improved volcanic eruption forecasting [69]. The integrated framework for these autonomous disaster response systems, illustrated in Figure 11, demonstrates the coordination of multiple hardware and software components for comprehensive emergency management.

6.4. Application Maturity Analysis

Applying the AIMM framework to drone applications reveals strategic patterns in technological development and deployment readiness, as systematically categorized in Table 6. The analysis demonstrates that applications with high deployment readiness (DR-4) typically operate in structured or semi-structured environments (AC-2/AC-3) with moderate autonomy levels. It reflects the current technological sweet spot for commercial viability (AL-2/AL-3). Applications requiring high intelligence integration (II-4), such as search and rescue and urban surveillance, show lower deployment readiness despite significant research investment. This specifically illustrates the challenges in transitioning advanced AI capabilities to operational systems. Medical delivery represents a successful case study in balancing capability and reliability. They have specifically achieved high deployment readiness through the implementation

Table 6

Application Maturity Assessment Using AIMM Framework

Application	AL	II	AC	DR
Medical Delivery (Zipline)	3	2	3	4
Last-Mile Package Delivery	3	3	4	2
Precision Agriculture (Mapping)	3	3	2	4
Precision Agriculture (Spraying)	2	2	3	3
Wildlife Monitoring	3	3	2	4
Infrastructure Inspection	2	3	3	4
Search and Rescue	2	4	4	3
Disaster Assessment	3	3	3	3
Urban Security Surveillance	2	4	4	2

of reliable autonomy with relatively straightforward intelligence requirements. This maturity assessment provides valuable insights for prioritizing development efforts and resource allocation across different application domains. This further emphasize the importance of carefully matching technological ambition with practical implementation constraints.

7. Comparative Analysis and Research Gaps

This section provides a critical synthesis and comparative analysis of research on AI and ACS in drones. It especially examines methodological approaches, performance metrics, and implementation challenges across diverse application domains. Here, we analyze primary research directly to identify patterns, contradictions, and critical gaps that inform both current understanding and future research directions while moving beyond mere summarization of existing literature. Our analysis reveals fundamental trade-offs between accuracy, robustness, and deployment readiness that characterize the current state of drone autonomy technologies.

7.1. Trends and Emerging Methodological Shifts (2015–2025)

Across the surveyed literature, three intertwined trends are reshaping autonomous drone research: (i) the migration from hand-crafted perception and linearized control toward end-to-end or tightly coupled perception–planning–control learning stacks, especially for vision-based autonomy; (ii) the shift from single-agent autonomy to cooperative, multi-agent decision making (e.g., MARL) for coverage, formation, and task allocation; and (iii) the move from cloud-centric pipelines toward onboard/edge AI, driven by latency, bandwidth, and resilience constraints. Recent surveys underscore that vision-based learning has become a core pathway to agility and autonomy, but remains bottlenecked by dataset realism, simulator fidelity, inference speed, and sim-to-real transfer [70]. In parallel, MARL-driven UAV control is rapidly expanding, yet evaluation practices remain fragmented and often incomparable across domains [71]. For multi-UAV swarms, decentralized coordination and robustness to communication limits are increasingly emphasized, with a growing need for scalable verification and safety-aware coordination [72].

7.2. Pitfalls, Failure Modes, and Reproducibility Bottlenecks

A cross-cutting finding is that reported performance gains frequently do not translate to operational robustness. Common pitfalls include: (a) *benchmark leakage* where models are implicitly tuned to narrow simulators or datasets; (b) *metric mismatch* where success is reported via task-specific scores that obscure safety-relevant failure modes (near-collisions, loss of control authority, latency spikes); (c) *sim-to-real brittleness* due to unmodeled aerodynamics, sensor artifacts, or domain shift; and (d) *incomplete ablations* that prevent attribution of improvements to specific modules. These pitfalls motivate a methodological shift toward standardized evaluation suites, explicit uncertainty reporting, and safety-constrained learning/control that remains stable under disturbances and degraded sensing. s

7.3. Performance Benchmarking Across Domains

A systematic comparison of control approaches based on primary research reveals fundamental trade-offs between accuracy, robustness, and deployment maturity, as comprehensively detailed in Table 7. This analysis demonstrates a clear accuracy-maturity trade-off where learning-based approaches achieve centimeter-scale tracking performance but exhibit low technology readiness levels (TRL 3-5). On the other hand, mature PID implementations offer proven

Table 7

Quantitative Benchmarking of Drone Control Approaches

Control Approach	Tracking Accuracy (RMSE)	Disturbance Rejection	Computational Efficiency	Deployment Maturity (TRL)
PID Control	10-15 cm	30-50% error increase	0.1-0.5 ms/iteration	High (9)
Model Predictive Control	3-8 cm	15-25% error increase	10-100 ms/iteration	Medium (6-7)
H_∞ Control	5-12 cm	10-20% error increase	2-5 ms/iteration	Medium (6)
Neural Network Control	4-10 cm	20-40% error increase	3-8 ms/iteration	Low-Medium (4-6)
Reinforcement Learning	2-6 cm	15-30% error increase	2-5 ms/iteration	Low (3-5)
Hybrid Hierarchical Control	3-7 cm	10-20% error increase	5-15 ms/iteration	Medium (5-7)

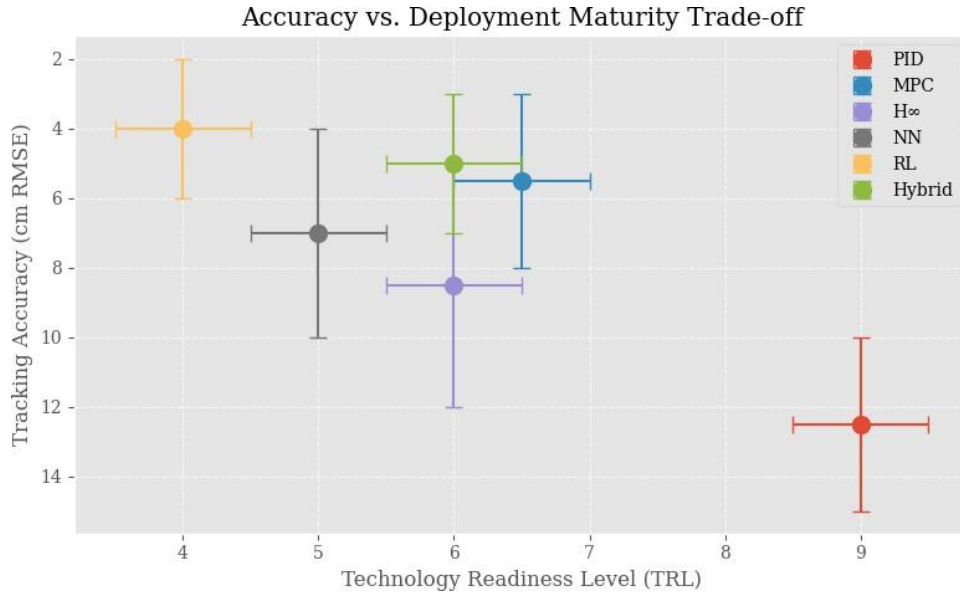


Fig. 12: Technology Readiness Level (TRL) versus tracking accuracy for control approaches. The inverse relationship highlights the implementation gap for advanced methods. Error bars show ranges from primary studies.

reliability at the cost of reduced precision (TRL 9). The inverse relationship between technological sophistication and deployment readiness, visually represented in Figure 12. This figure shows that there is an implementation gap that currently separates advanced control methodologies from field-deployable systems. This pattern displays the challenges in transitioning from laboratory demonstrations to operational deployments, particularly for methods that require extensive training data or exhibit sensitivity to environmental variations.

Perception systems exhibit similarly complex performance characteristics that vary significantly across operational contexts, with no single approach dominating all conditions, as detailed in Table 8. The context-dependent nature of perception performance necessitates careful application-specific selection. Thermal imaging specifically demonstrates particular effectiveness for search and rescue operations in darkness. On the other hand, LiDAR-camera fusion excels in infrastructure inspection applications, while event-based vision provides advantages for agile flight scenarios with high dynamics. Radar-vision fusion also enables reliable all-weather operations despite higher computational requirements. This specialization shows the importance of matching perception technologies to specific operational requirements rather than seeking universally optimal solutions.

7.4. AIMM Framework Synthesis and Research Gaps

Applying the AIMM framework across the considered studies reveals critical patterns in the research landscape that illuminate both current concentrations and significant opportunities. Quantitative analysis demonstrates imbalance in research focus, with the majority of published studies concentrating on systems exhibiting high intelligence integration (II-3/II-4) but low deployment readiness (DR-1/DR-2). On the other hand, only a small fraction addresses the moderate intelligence integration with high deployment readiness that characterizes most commercially successful applications.

Table 8

Cross-Domain Performance Analysis of Perception Systems

Perception Approach	Object Detection Accuracy	Performance in Challenging Conditions	Computational Requirements	Environmental Robustness
CNN-based Detection	85-92% mAP	40-60% accuracy in low light	8-20 GFLOPS	Moderate
Event-based Vision	75-85% mAP	70-80% accuracy in high dynamics	2-5 GFLOPS	High
LiDAR-Camera Fusion	88-94% mAP	75-85% accuracy in adverse weather	15-30 GFLOPS	High
Thermal Imaging	80-88% mAP	85-90% accuracy in darkness	5-10 GFLOPS	Temperature-dependent
Radar-Vision Fusion	82-90% mAP	80-90% accuracy in adverse weather	12-25 GFLOPS	Very High

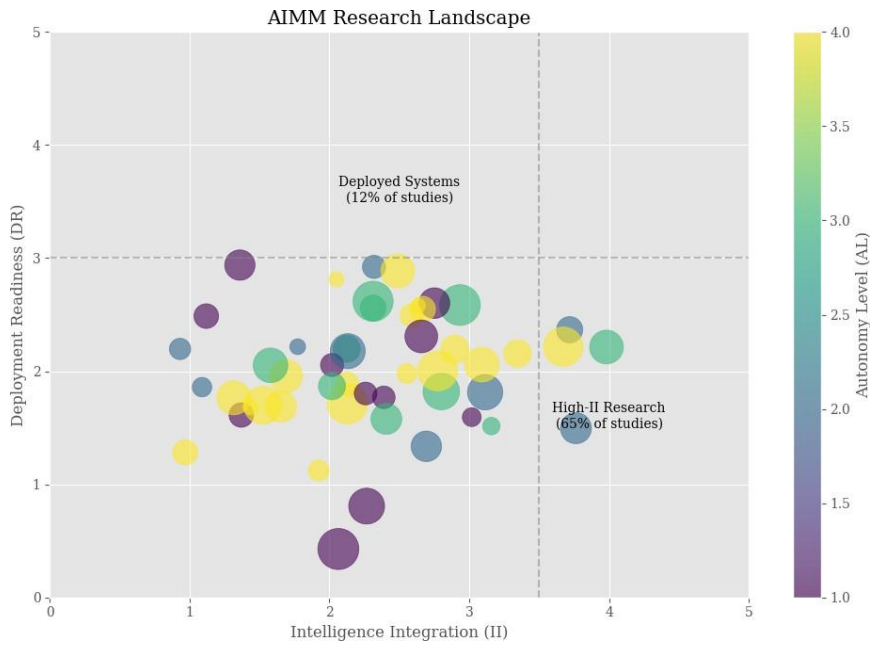


Fig. 13: AIMM framework visualization showing research concentration (bubble size) and performance gaps. The x-axis shows II, y-axis DR, and bubble color represents Autonomy Level.

This distribution reveals a significant gap between academic research priorities and commercial implementation realities. Systems designed for high application complexity environments consistently show lower deployment readiness compared to those targeting moderate complexity scenarios. This further reveals important challenges in achieving reliable operation in unstructured, dynamic settings. The visualization of this clustering in Figure 13 clearly illustrates the concentration of research activity in high-intelligence, low-readiness regions compared to the operational systems that cluster in moderate-intelligence, high-readiness areas.

The dimensional analysis identifies several underexplored but strategically promising combinations that represent viable pathways for advancing the field. Systems combining high autonomy with moderate complexity (AL-4, AC-2/AC-3) present opportunities for near-term deployment of advanced capabilities without confronting the challenges of fully unstructured environments. Similarly, the moderate intelligence with high deployment readiness combination (II-2/II-3, DR-4), despite representing the profile of most commercial successes, receives disproportionately little research attention. Thus, this represents an important research gap that requires more investments. Perhaps most critically, there exists a pressing need for bridging technologies that can transition high-intelligence systems from low to high deployment readiness through focused attention on reliability, verification, and operational simplicity. This structured gap analysis provides valuable strategic guidance for prioritizing research investments to maximize practical impact.

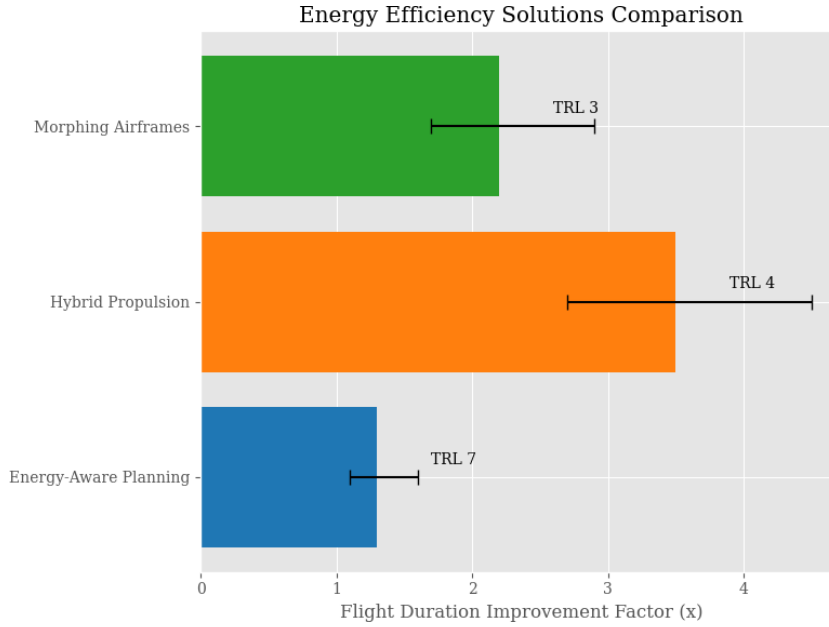


Fig. 14: Comparative analysis of energy efficiency solutions showing potential flight duration improvements versus technological readiness. Error bars indicate performance ranges from experimental studies.

and systematically accelerate the transition from laboratory demonstrations to field-deployable systems that can deliver tangible societal benefits.

8. Challenges and Future Directions

The integration of AI and ACS in drones, while promising, faces significant technical, regulatory, and ethical challenges that must be addressed to enable widespread deployment. This section categorizes these challenges and identifies promising research directions to overcome them.

8.1. Technical Challenges

Energy limitations represent a fundamental constraint for drone operations, with current battery technologies restricting flight durations and payload capacities across most commercial platforms [73, 74]. Research into energy-aware trajectory planning has demonstrated good improvements in energy consumption through optimized path and velocity selection [75, 76, 77]. Hybrid propulsion systems combining batteries with alternative power sources show particular promise for extending operational endurance, while morphing airframe designs offer more modest but meaningful gains in aerodynamic efficiency [78, 79]. The comparative potential of these energy solutions is illustrated in Figure 14, which compares the potential flight duration improvements and technological readiness of these approaches.

Operational robustness in adverse weather conditions remains another critical challenge, as drones are particularly sensitive to environmental factors such as precipitation, high winds, and temperature extremes [80]. Multi-modal sensing approaches have emerged as essential for maintaining reliable performance under challenging conditions, with sensor fusion strategies demonstrating better resilience compared to single-modality systems. For instance, LiDAR-camera fusion maintains high detection capability in poor visibility conditions where vision-only systems experience considerable degradation. The relative performance of different approaches under adverse conditions is quantitatively compared in Table 9.

Safety assurance for learning-based systems presents unique verification challenges due to the inherent complexity and black-box nature of many AI algorithms [81]. The formal verification of neural network controllers remains particularly difficult. Therefore, research into alternative approaches should be initiated. This might include structurally

Table 9
Adverse Condition Performance Comparison

Approach	Wind	Precipitation	Temperature
Standard Systems	65%	50%	60%
All-Weather Perception	85%	85%	75%
Adaptive Control	90%	70%	80%
Resilient Hardware	75%	90%	95%

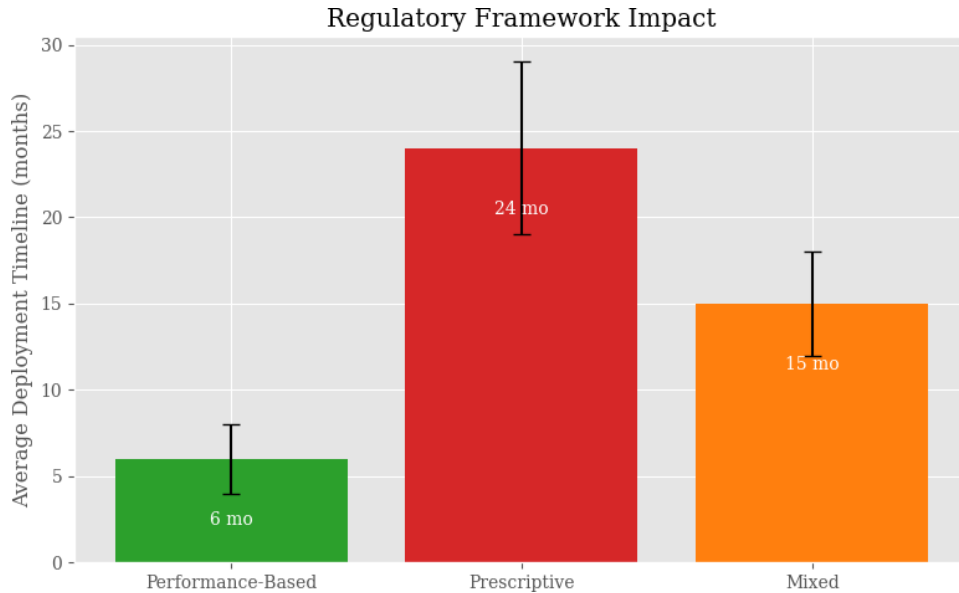


Fig. 15: Impact of regulatory frameworks on deployment timelines across regions. Performance-based approaches enable 3-5x faster deployment. Error bars show variation across application domains.

constrained network architectures that facilitate verification while maintaining performance, runtime monitoring frameworks for detecting unsafe states during operation, and hybrid system designs that combine learning-based components with verifiable safety layers [82, 83].

Computational efficiency represents another barrier, as the extra processing demands of advanced AI algorithms often exceed the capabilities of resource-constrained drone platforms. Model compression techniques have demonstrated high effectiveness in reducing computational requirements while preserving accuracy. Also, specialized hardware approaches, including neuromorphic computing and application-specific integrated circuits, offer promising pathways for achieving substantial improvements in energy efficiency for specific computational workloads [84, 85, 86].

Reliable navigation in GPS-denied environments remains an active research area, with visual-inertial odometry (VIO) and simultaneous localization and mapping (SLAM) serving as foundational approaches. However, these methods face limitations in textureless environments and suffer from accumulated drift over extended operations. Recent research integrating deep learning with traditional state estimation has shown promising directions for improving robustness and accuracy in challenging navigation scenarios [87, 88].

8.2. Regulatory and Ethical Considerations

The regulatory landscape for drone operations continues to evolve, with a discernible trend toward performance-based frameworks that enable advanced operations under appropriate safety cases. Regulatory models such as EASA's risk-based categorization have demonstrated effectiveness in streamlining operations while maintaining safety standards. Nevertheless, there are regulatory challenges, particularly for beyond visual line of sight operations and the certification of learning-based systems, where traditional deterministic verification approaches prove inadequate

[89, 90]. The impact of different regulatory frameworks on deployment timelines is illustrated in Figure 15. The figure shows the advantages of performance-based approaches.

Ethical considerations and privacy concerns substantially influence public acceptance of drone technologies. Surveys consistently indicate significant public apprehension regarding privacy implications, particularly concerning facial recognition capabilities and persistent monitoring applications. Algorithmic bias represents another critical ethical challenge, with documented disparities in performance across different demographic groups. Addressing these concerns requires comprehensive approaches. These approaches include privacy-by-design methodologies that incorporate protections into system architectures, algorithmic impact assessments to identify potential biases before deployment, and enhanced transparency and explainability to help users understand system decisions [91, 92].

8.3. Promising Research Directions

Algorithmic advances continue to drive progress across multiple domains of drone autonomy. Neuro-symbolic AI approaches that combine neural networks with symbolic reasoning show considerable promise for improving interpretability and data efficiency. User studies have specifically demonstrated better operator understanding compared to black-box systems [93, 94]. Meta-learning techniques enable rapid adaptation to new tasks with minimal training examples. They have demonstrated good performance in few-shot learning scenarios. Uncertainty-aware decision making represents another promising direction, with research showing further improvements in safety through explicit reasoning about perceptual and predictive uncertainties [95].

Hardware-software co-design approaches offer high potential for overcoming computational and energy limitations. Specialized processors designed specifically for drone navigation tasks demonstrate notable improvements in energy efficiency compared to general-purpose computing platforms. Sensor-algorithm co-design enables novel capabilities such as high-speed navigation using event-based vision systems that would be challenging with conventional approaches. This algorithm involves the joint optimization of sensing hardware and processing algorithms [85, 96].

Research in human-AI teaming focuses on developing effective collaboration frameworks between human operators and autonomous systems. Well-designed human-AI teams demonstrate measurable performance improvements, while shared mental models significantly reduce operator errors in complex scenarios [97, 98]. Bio-inspired approaches offer additional promising directions, with insect-inspired control mechanisms enabling exceptionally low-latency flight stabilization and flocking behaviors derived from bird colonies providing robust coordination strategies for drone swarms [99, 100, 101].

8.4. Roadmap for Bridging the Research-Deployment Gap

A systematic development and testing approach is essential for translating research advances into operational systems. High-fidelity simulation environments have demonstrated the capability to closely approximate real-world performance while accelerating development cycles [102]. Structured testing frameworks are more efficient than unstructured approaches. This enables comprehensive failure mode identification in considerably less time. Incremental deployment strategies have shown improvements in achieving operational readiness while reducing critical incidents during implementation [103].

Strategic partnerships between industry and academia serve as a critical catalyst for technology advancement. Collaborative initiatives built around open benchmarking platforms have demonstrated accelerated development timelines. Also, shared testbed facilities substantially compress the transition from laboratory development to field testing. Formal collaborative research programs have consistently yielded higher rates of technology commercialization and market adoption [104, 105].

The establishment of comprehensive standardization frameworks represents another essential component for bridging the deployment gap. Significant variations in performance evaluation methodologies across different research groups highlight the necessity for standardized metrics and testing protocols [106]. The development of standardized safety cases has proven effective in streamlining regulatory approval processes, while interoperability standards reduce integration complexity and accelerate deployment timelines [107].

By addressing technical challenges, ethical considerations, and implementation barriers through these structured approaches, the field can systematically accelerate the transition from laboratory demonstrations to widespread deployment of AI-enabled drone systems. This coordinated effort will enable the realization of significant societal benefits while ensuring appropriate risk management throughout the technology adoption lifecycle.

9. Conclusion

This review re-examined the literature on AI-enabled drones through a structured synthesis of autonomous control, perception, navigation, decision-making, and application evidence. Rather than treating the field as a uniform progression toward full autonomy, the analysis shows a more uneven landscape in which algorithmic progress is substantial but translational maturity varies widely across domains and methodological families. The most robust near-term advances are found in systems that combine bounded autonomy, multi-modal sensing, and carefully engineered hybrid architectures, whereas claims of broad autonomous capability often remain supported mainly by simulation or tightly controlled experiments.

A central contribution of the manuscript is the use of the AIMM framework to distinguish technical sophistication from deployment readiness. Across the reviewed studies, this distinction clarifies why high-performing methods do not automatically translate into deployable systems. The literature repeatedly points to the same bottlenecks: limited field validation, weak comparability of reported metrics, incomplete treatment of failure modes, energy and compute constraints, and unresolved safety and assurance issues for learning-enabled autonomy. These are not peripheral limitations; they are the principal reasons why the research-to-deployment gap remains a defining characteristic of the field.

The review also indicates that future progress is most likely to come from research programs that integrate methodological rigor with deployment realism. This includes stronger evaluation protocols, clearer reporting of operational context, more explicit treatment of uncertainty and failure, and closer alignment between AI innovation and the constraints of real airborne systems. From an academic standpoint, the field will benefit from moving beyond isolated performance claims toward more comparable, reproducible, and system-level evidence. From an applied standpoint, the most credible pathway forward lies in application-specific autonomy, staged deployment, and hybrid architectures that balance adaptability with verifiability.

Accordingly, the main takeaway is not that AI-enabled drones are nearing universal autonomous deployment, but that the field now has a sufficiently mature evidence base to define where progress is genuine, where claims should be narrowed, and where future work can produce the greatest scientific and operational value. Framed in this way, the manuscript contributes a more disciplined and academically grounded account of the present state of AI-enabled drone systems and the priorities required to strengthen their next phase of development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Compliance with Ethics Requirements

No dataset was generated by the present review. The manuscript synthesizes published literature and does not report new experimental data. Any quantitative statements discussed in the text are drawn from the cited sources and should be interpreted within the scope and conditions reported by those original studies. AI assistance was used only for grammar correction and language polishing.

Research Context

This work was conducted within the AEROBOTEK research initiative at Kocaeli University, a multidisciplinary *R&D* team focused on autonomous aerial systems. Our hands-on experience with aerial robotics platforms and collaborations with industry partners informed the selection and analysis of trends, with emphasis on practical feasibility and implementation challenges.

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